

G04 Review A - PROTECTED

OPEN ONLY AFTER THE MATCHING GATE ATTEMPT HAS BEEN SAVED. This Review is diagnostic and does not provide a copyable implementation.

Objective control panel

Control	Expected
Dimension rows	5
Duplicate dimension keys	none
Base rows	11
Base valid	6
Base quarantined	5
Base quarantine IDs	S105, S106, S107, S108
Delta rows	3
Delta valid	3
Target rows after delta	8
Successful rows after delta	6
Successful value after delta	540.00
Checkpoint after good delta	2026-09-02T09:10:00
Failure batch invalid?	True
Checkpoint after failure	UNCHANGED_FROM_GOOD_DELTA
Target after failure	UNCHANGED_FROM_GOOD_DELTA
API raw / unique	5 / 4
API duplicate IDs	E3

Serving controls by group

Group	Count	Value
East	2	185.00
North	2	200.00
South	1	55.00
West	1	100.00

Critical diagnostic patterns

K1 - PostgreSQL, modeling, constraints, transactions and database safety

Wrong database/schema/role context; constraints absent or bypassed; transaction boundary cannot prove rollback; fact/dimension grain is unclear; duplicate dimension key causes fan-out.

K2 - File/API/database ingestion contracts, defensive extraction and schema handling

Raw files edited in place; malformed rows silently dropped; only first API page read; nested/typed fields coerced without evidence; source-to-accepted counts do not reconcile.

K3 - Incremental state, CDC, idempotency, late data, replay and atomic checkpoints

Checkpoint advances before successful commit; rerun duplicates rows; delta update order is undefined; duplicate event identity is double-counted; late data has no explicit rule; failed attempt overwrites evidence.

K4 - ETL/ELT transformations, quality tests, reconciliation and serving contracts

Serving table has no stated grain; dedup hides key defects with DISTINCT; quality checks are visual/manual only; source-to-target totals disagree; transformation output changes on identical input.

K5 - Docker images, containers, Dockerfile, ports, volumes, networks, Compose and secrets

Container works only through hidden host state; Dockerfile runs as root without reason; bind/named-volume behavior is misunderstood; Compose networking uses localhost incorrectly; secrets are committed/logged.

K6 - Airflow DAGs, schedules, retries, params, connections, sensors, backfills and debugging

DAG imports side effects; dependencies are implicit; schedule/data interval is confused with wall-clock execution; retries are unbounded; sensor waits forever; params/connections are hard-coded; backfill duplicates data; logs cannot locate the failed task.

K7 - Integrated operations: observability, recovery, reproducibility, evidence and technical defense

Run cannot reproduce from clean state; evidence is only screenshots with no controls; failed run is erased; runbook omits prerequisites; technical explanation cannot connect state/quality/runtime choices to observed results.

Targeted repair menu - changed micro-proof

Weak	Fresh repair proof
K1	Use a fresh 5-row table pair with a different grain. Recreate keys/constraint/transaction rollback and explain why one index or no index is justified.
K2	Use a new 6-row CSV plus two-page JSON payload. Quarantine one schema/business defect and prove pagination + count reconciliation without editing raw input.
K3	Use a changed timestamp field and two-record delta. Prove same-input rerun = no duplicate; inject a failed batch and prove checkpoint stays unchanged; include one late event.
K4	Build a new serving aggregate at a different group grain. Add not-null/unique/relationship/business-rule tests and an independent source-to-target total.
K5	Containerize a different tiny command. Prove configuration injection, a named volume or justified bind mount, service-name networking, non-root execution and no committed secret.
K6	Create a different 3-4 task DAG. Add one retry/timeout, parameter, explicit dependency and changed schedule/backfill case. Force one task failure and diagnose it from logs.
K7	Reproduce a changed mini-pipeline from an empty directory/clean clone and produce a new run record + 90-second explanation with the Review closed.

Routing after repair

Append repair evidence to the original attempt. Do not erase failures. If targeted repair now demonstrates all weak competencies, the Gate family may pass without exposing the next sealed Gate. Gate B/C are fresh full reassessments only when explicitly assigned.